

Names

Jackson Glass, Simone Minor, Daria Slowinska, JJ Sutkus

Question

Is a hot dog a sandwich?

Hypothesis

1/4 of UVA full-time undergraduate students believe a hot dog is a sandwich.

Pre-Analysis Steps

The dependent variable we are measuring is the percentage of full-time undergraduate students who believe a hot dog is a sandwich. This means that we have to clean the data to eliminate respondents who are not full-time undergraduate students. Our variable of interest is discrete because the overall percentage we are measuring is of binary responses: yes or no. First, we should visualize the data using frequency tables and bar charts to display the response count for each answer yes, affirming that a hot dog is a sandwich, and no, denying that a hot dog is a sandwich. We could also use a pie chart after cleaning the data to see all the responses from full-time undergraduate students, and what portion said yes and which said no.

Analysis Methods

We should do a proportion test like Chi-square or a Z-test to see if our respondent population of full-time undergraduate students is representative of the entire population by comparing the school and class proportion of the total population. A Chi-Square Test of independence compares our categorical data (e.g., year, college at UVA) frequencies in our sample against expected frequencies from the population to see if they match. We should also do a binomial test to see if the observed response proportion is the same as our expected response proportion.

Chi-Squared Test of Independence Equation:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

χ^2 = the test statistic $\sum$ = the sum of

O = Observed frequencies E = Expected frequencies

Evaluation of Success

After running the chi-square test of independence or the Z-test (ultimately testing the same thing, proportion accuracy), we will need to measure success of the test in relation to our hypothesis. We will determine if we can reject or affirm our hypothesis using the p-value. If the calculated p-value is less than or equal to 0.05, then we will reject the null hypothesis because observed data will significantly differ from random chance. However, if it is greater than 0.05 then we will fail to reject the null hypothesis (meaning we accept it) because there is no significant difference between our hypothesized ¼ of UVA full-time undergraduate students that we thought would believe a hot dog is a sandwich and those who actually did.